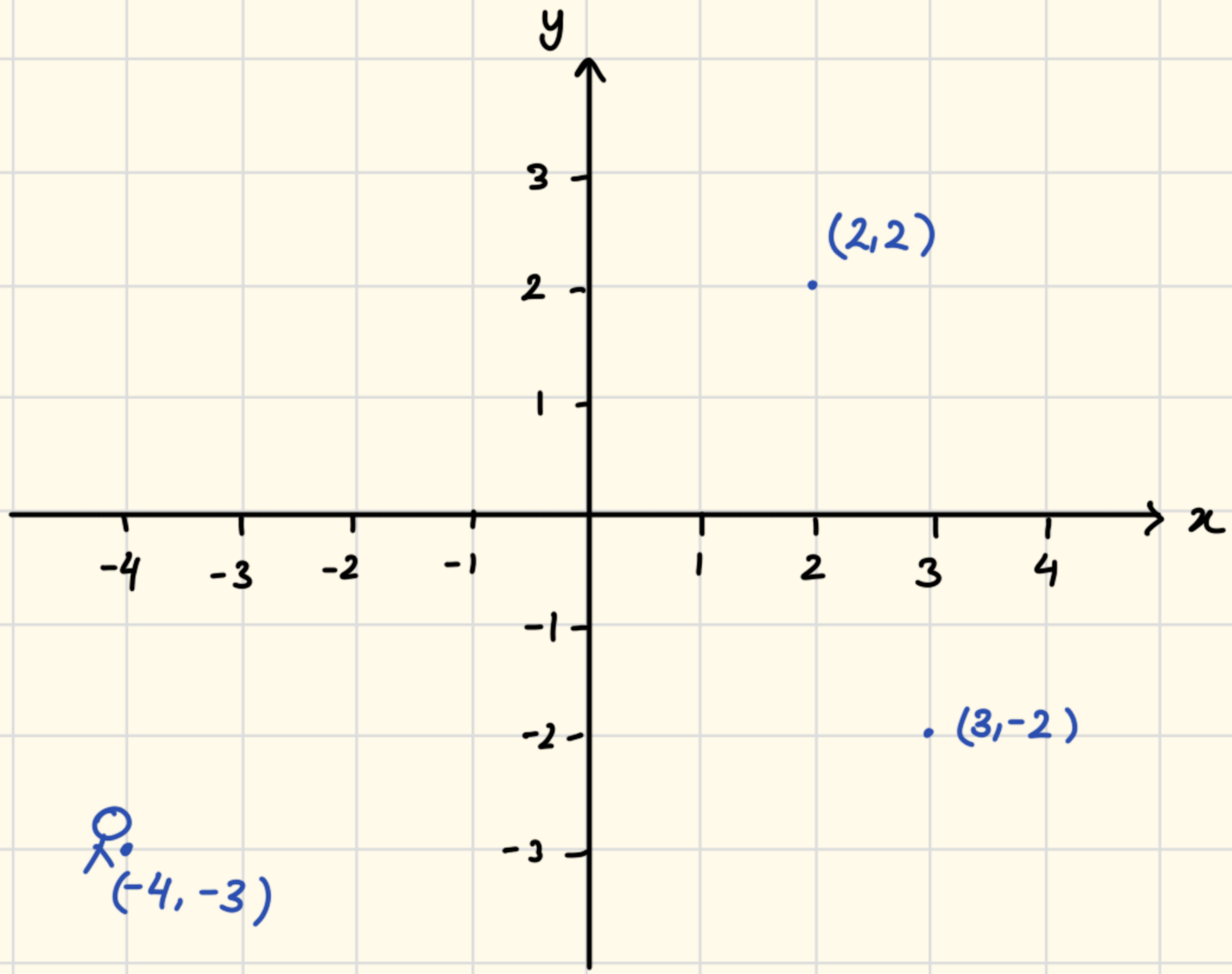
Coordinates 8

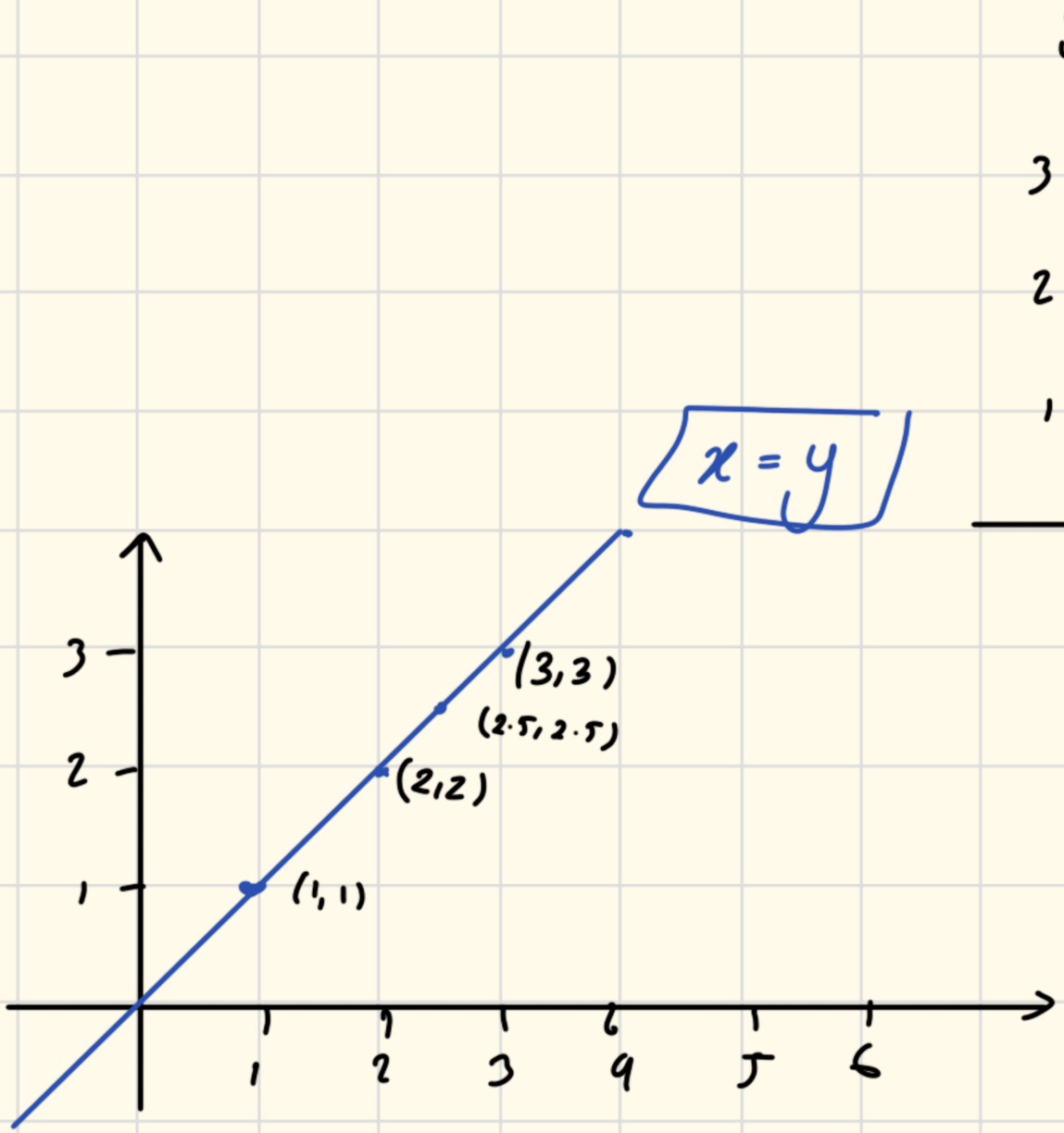
Represent locations in space.

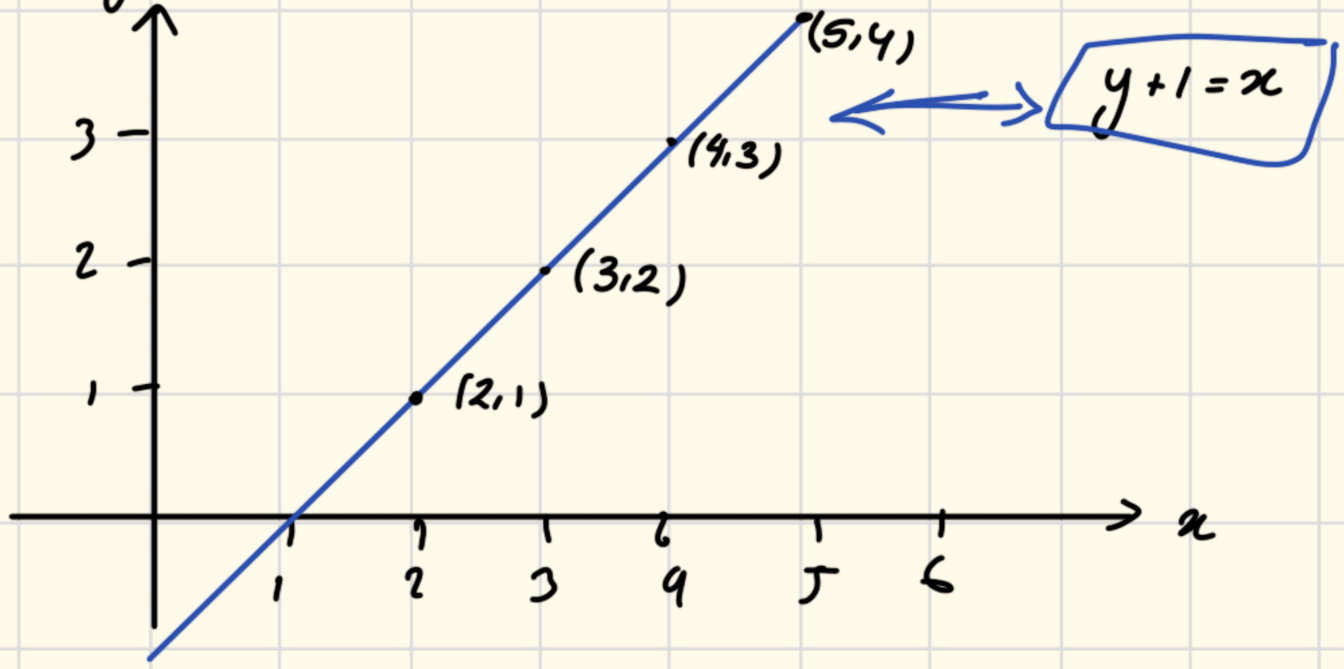


(x, y) always x then y

LINES

Our goal is to find a relationship between x & y!





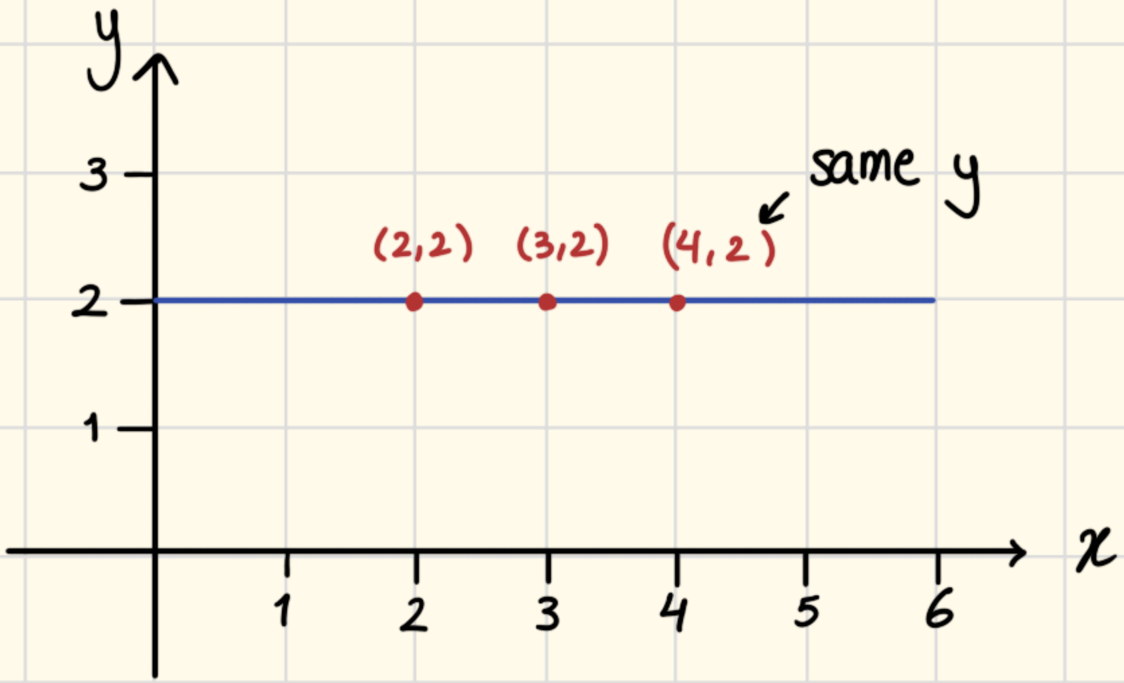
Horizontal And Vertical Lines

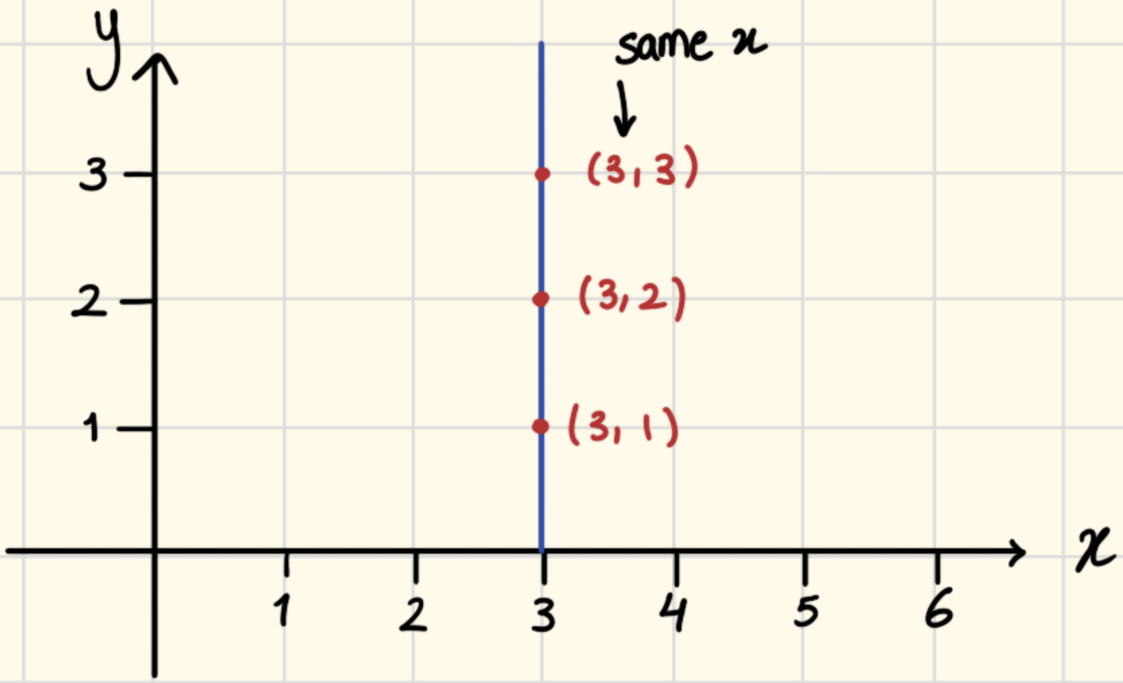
Horizontal

$y =$

Vertical

$x =$



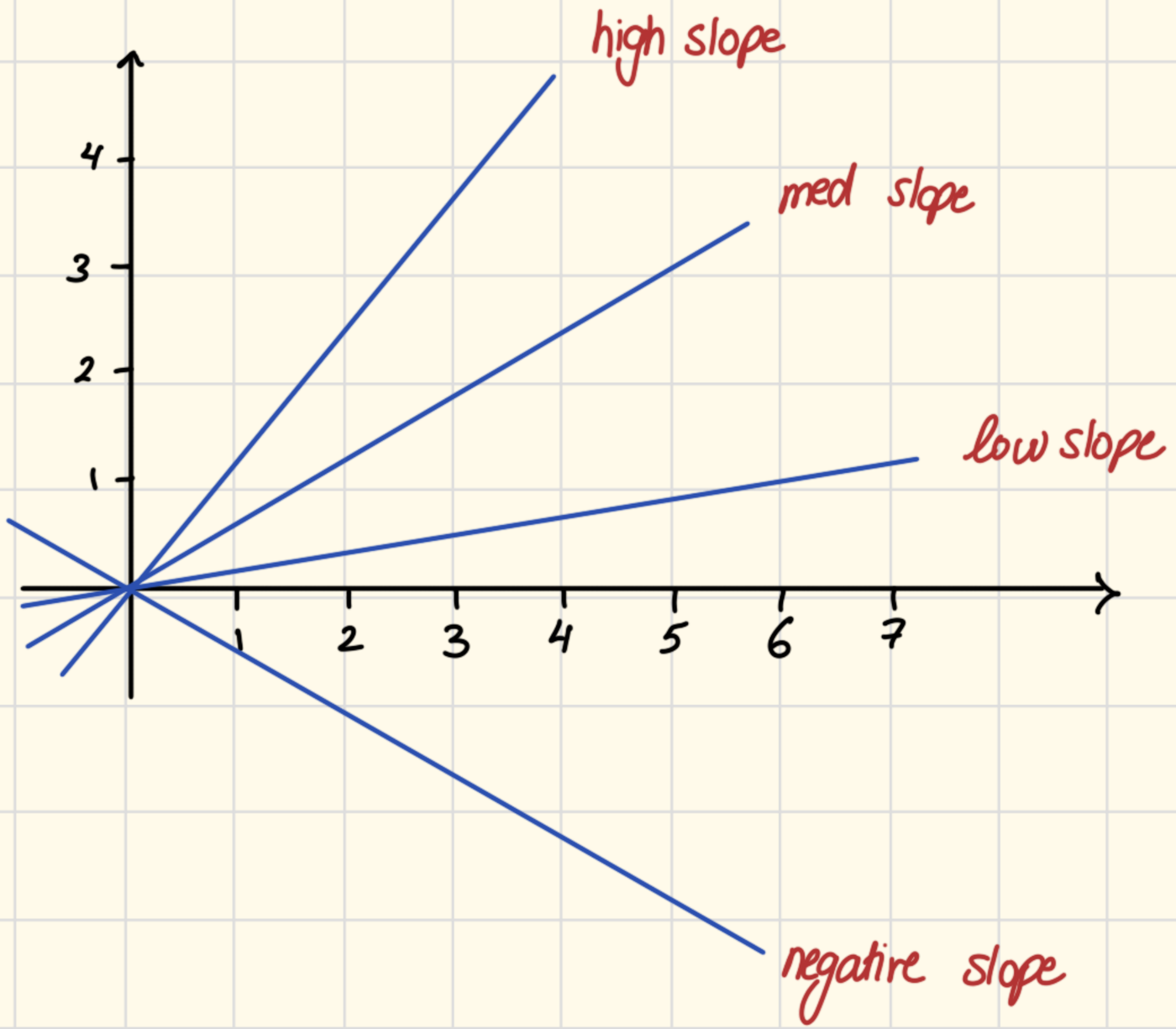


$y = 2$

$x = 3$

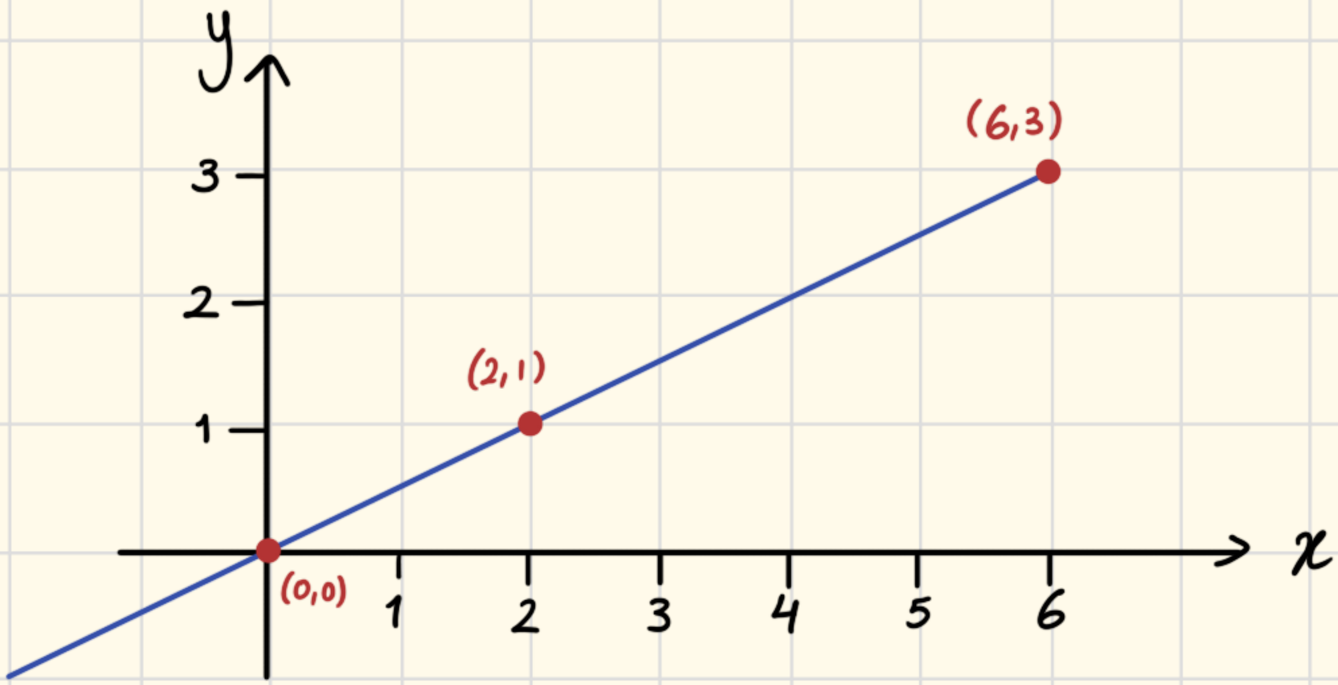
Slope / Giradiant

Measures steepness of line



SLOPE / GRADIANT

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$



$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{2 - 0} = \frac{1}{2}$$

$$= \frac{3 - 1}{6 - 2} = \frac{2}{4} = \frac{1}{2}$$

* You can use any 2 points !

$$H, (\overset{x}{6}, \overset{y}{6})$$

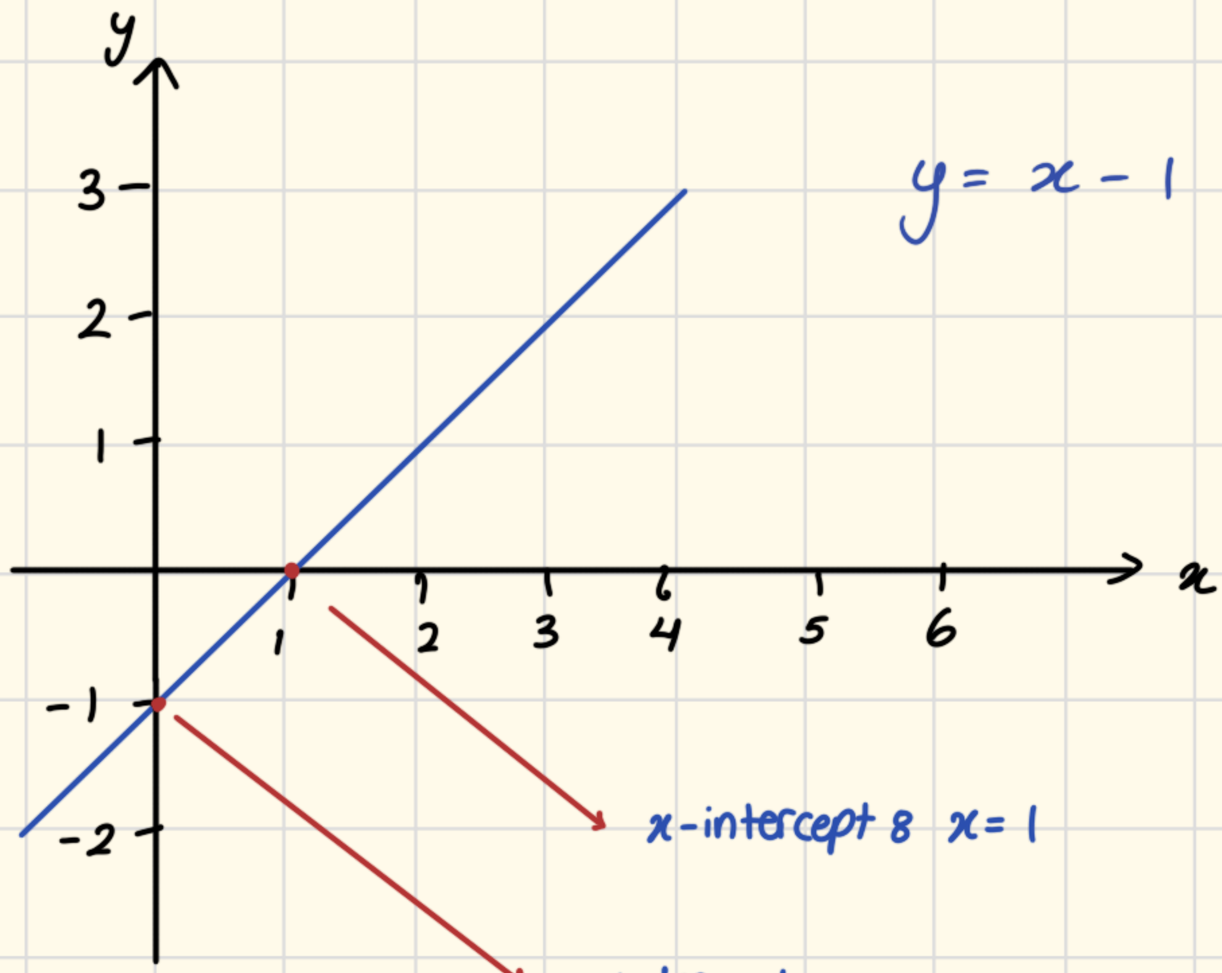
$$G, (\overset{x}{2}, \overset{y}{-5})$$

$$\frac{6-(-5)}{6-2} = \frac{11}{4}$$

Reem

INTERCEPTS

Are the intersection points of a line with x, y axis.



$y = x - 1$

x-intercept & $x = 1$

y-intercept & $y = -1$

Equation Of Line

You need the slope and y-intercept

$$y = mx + c$$

m ↓ slope

c ↓ y-intercept

Equation Of Line

You need the slope and point at line.

$$y - y_0 = m(x - x_0)$$

m ↓ slope

x_0 ↓ point on line

Equation Of Line

Find the equation of line with 8

a) Slope = 2, y-intercept = 3

b) Slope = 2, point (0,4)

c) Slope = $\frac{1}{3}$, point (6,4)

d) point (3, 2), point (6,4)

Equation Of Line

a) $y = mx + c \Rightarrow y = 2x + 3$

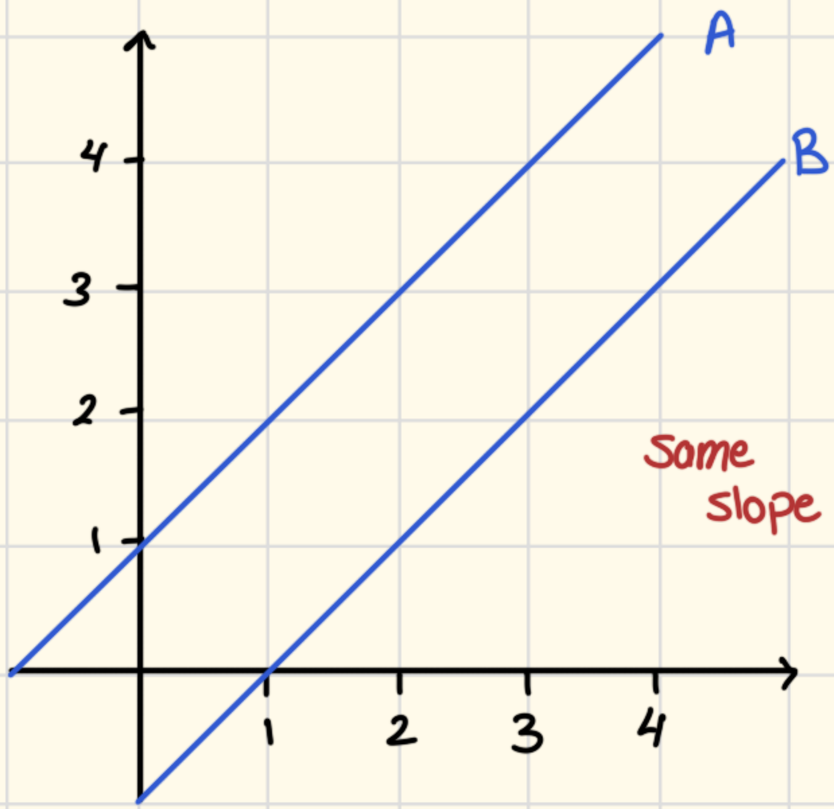
b) $y = mx + c \Rightarrow y = 2x + 4$

c) $y - y_0 = m(x - x_0) \Rightarrow y - 4 = \frac{1}{3}(x - 6)$

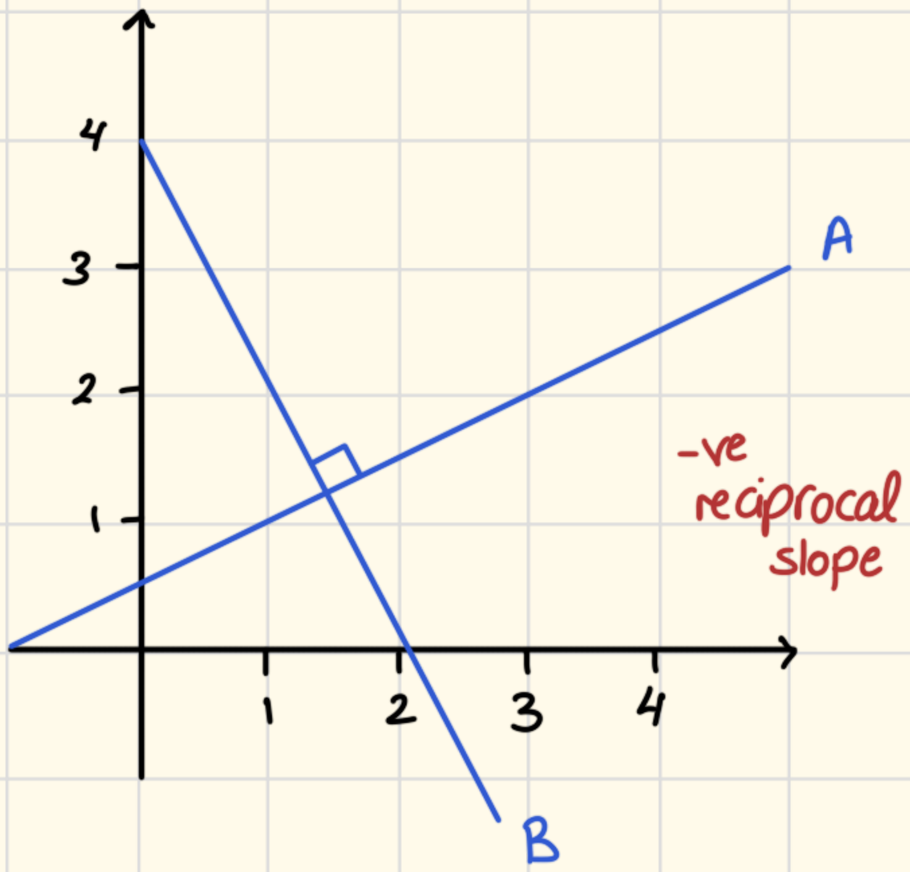
d) $y - y_0 = m(x - x_0) \Rightarrow m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 2}{6 - 3} = \frac{2}{3}$

$$y - 4 = \frac{2}{3}(x - 6)$$

Parallel And Perpendicular Lines



$$m_A = m_B$$



$$m_A = -\frac{1}{m_B}$$

Equation Of Line

Line $\mathcal{L}$: $2x + 3y = 6$

Find the equation of line R that is :

a) parallel to $\mathcal{L}$ and passes point $(-1, 3)$

b) perpendicular to $\mathcal{L}$ and passes point $(-1, 3)$

Equation Of Line

Line $\mathcal{L}$: $2x + 3y = 6$

Find $m_{\mathcal{L}}$. $y = \underbrace{-\frac{2}{3}}_{m_{\mathcal{L}}} x + \frac{6}{3}$

a) Since $\mathcal{L} \parallel R$, $m_R = m_{\mathcal{L}} = -\frac{2}{3}$

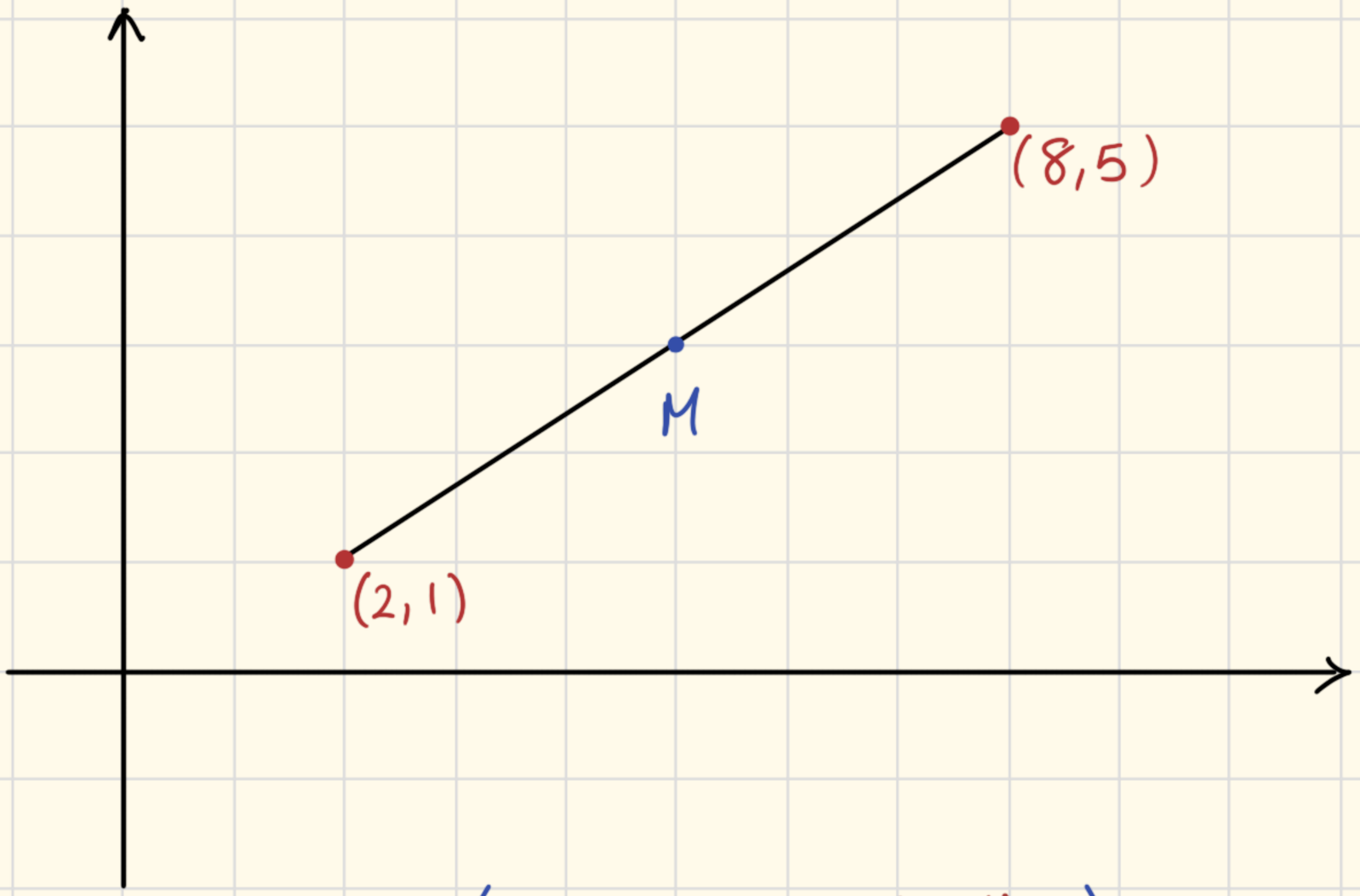
$$y - 3 = -\frac{2}{3}(x + 1)$$

a) Since $\mathcal{L} \perp R$, $m_R = -\frac{1}{m_{\mathcal{L}}} = \frac{3}{2}$

$$y - 3 = \frac{3}{2}(x + 1)$$

MIDPOINT

A midpoint is a point between two points.



$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

QUADRATIC GRAPHS

The graph of a quadratic is called a parabola.

$$ax^2 + bx + c = 0$$

POSITIVE a



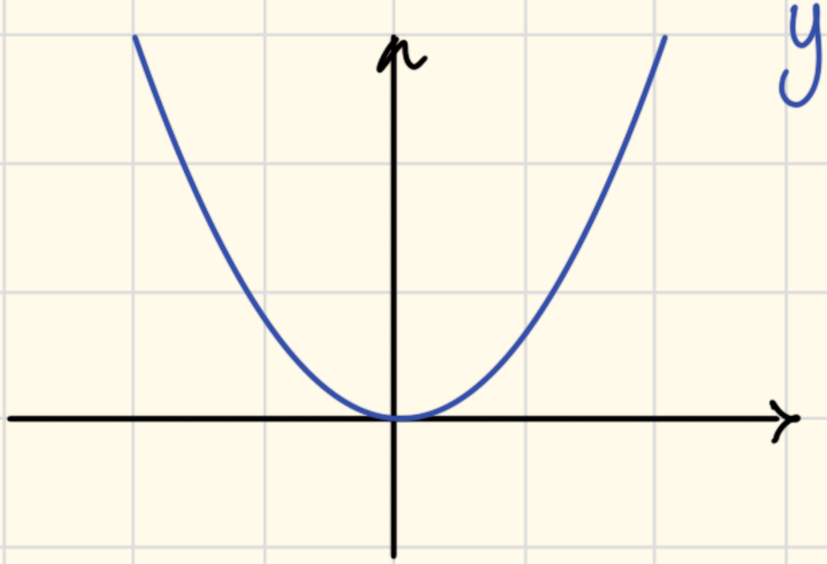
Smile

NEGATIVE a

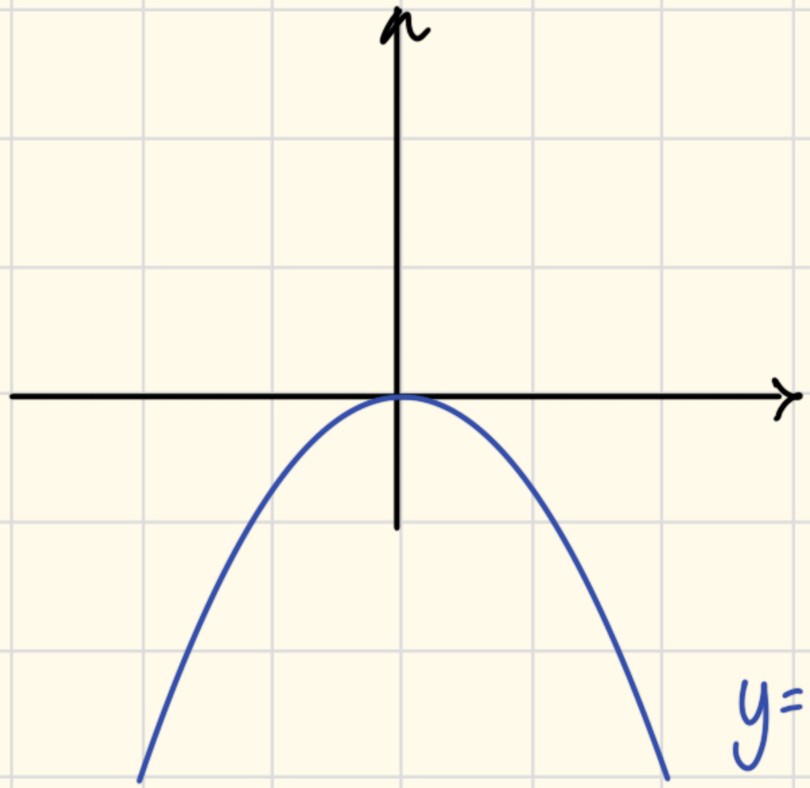


Frown

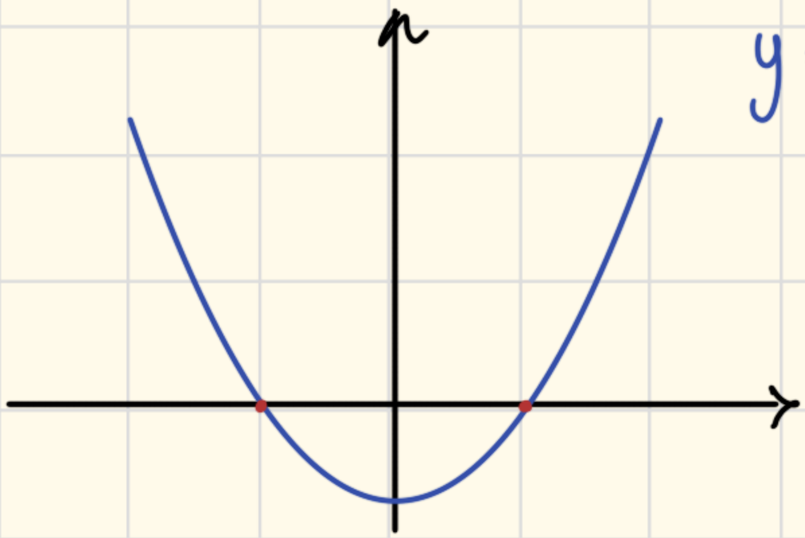
QUADRATIC GRAPHS



$y = x^2$

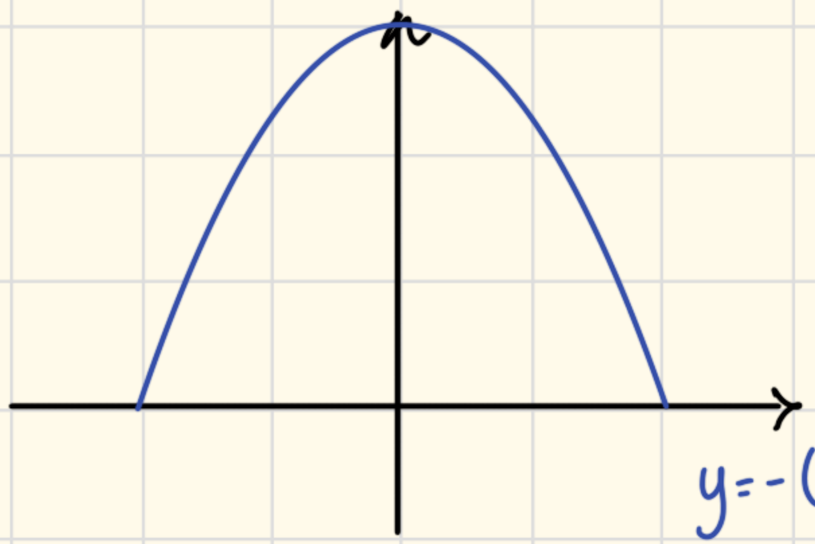


$y = -x^2$



$y = (x+1)(x-1)$

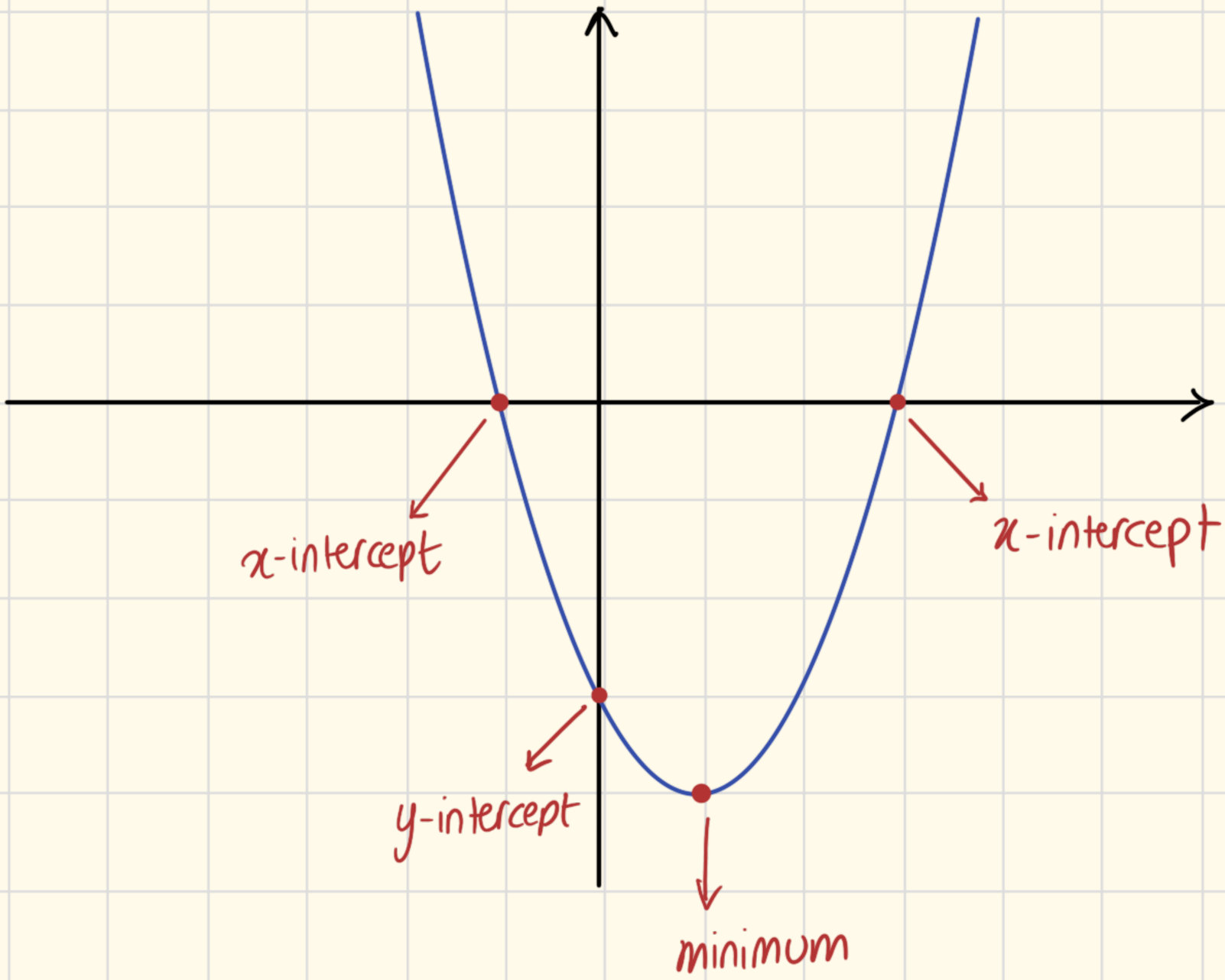
$x^2 - 1$



$y = -(x+2)(x-2)$

$= -x^2 + 4$

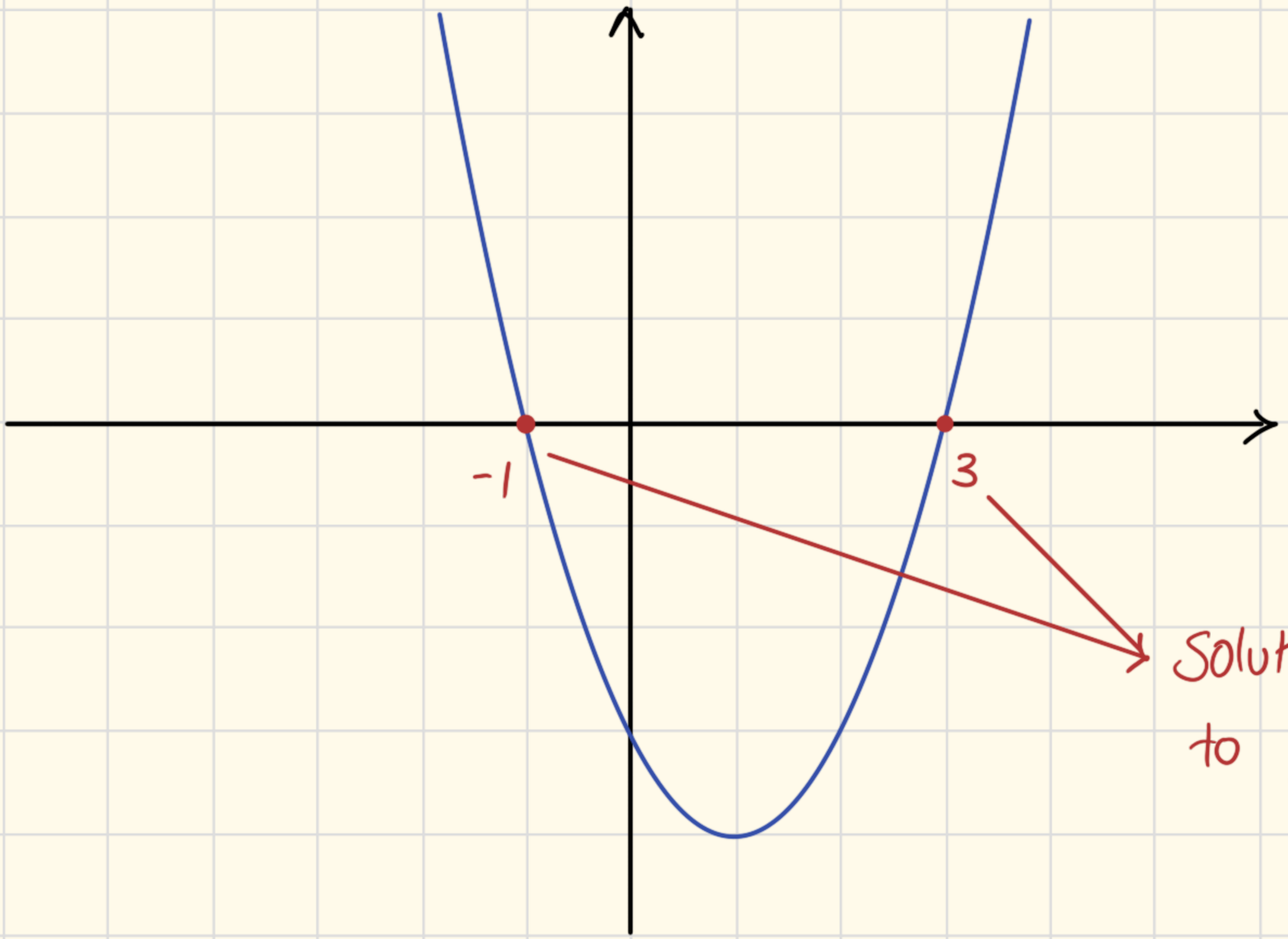
QUADRATIC GRAPHS



QUADRATIC GRAPHS

$$y = (x - x_1)(x - x_2)$$
$$= (x + \underline{1})(x - \underline{3})$$

x_1, x_2 solutions & x-intercepts

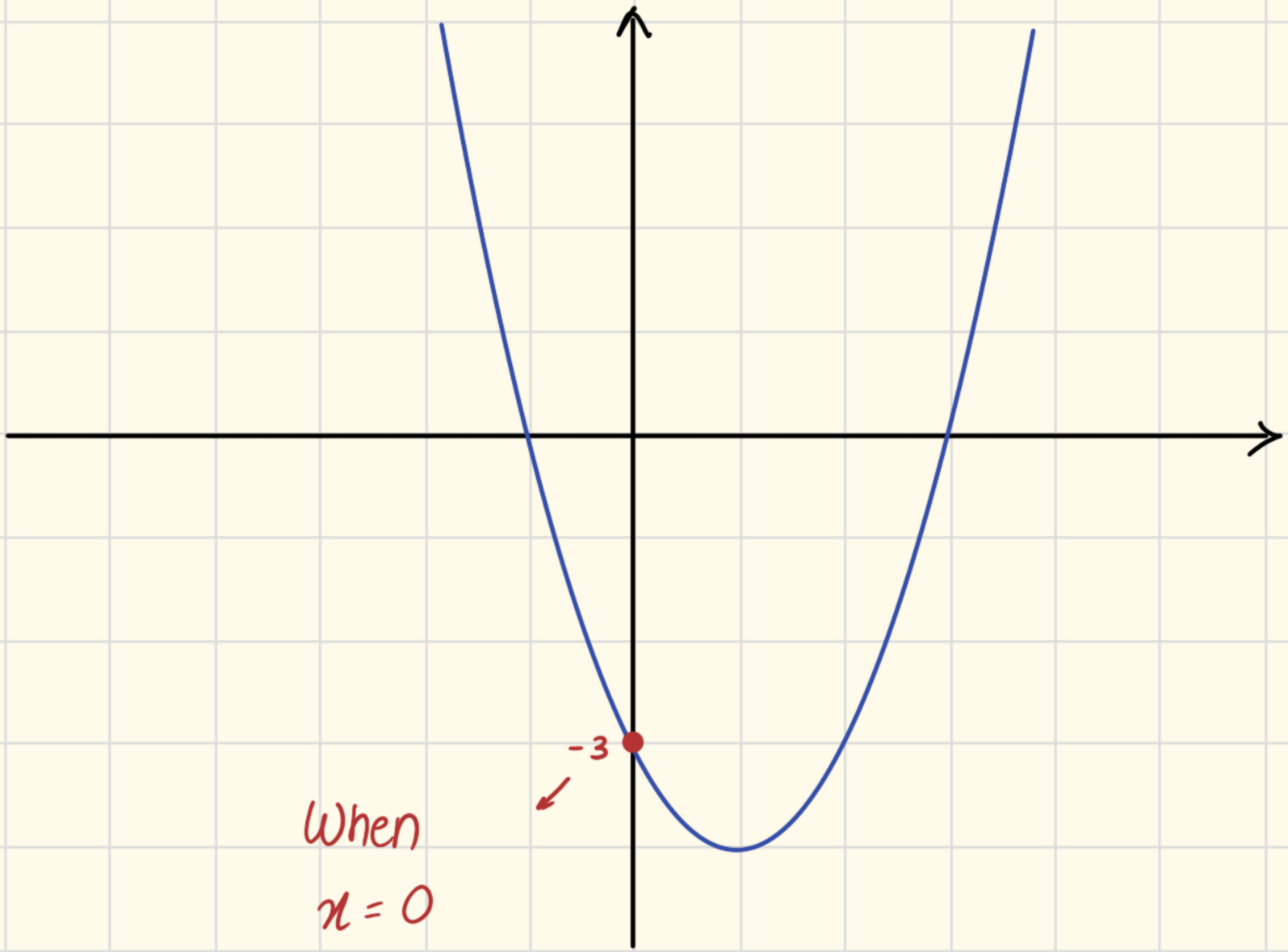


QUADRATIC GRAPHS

$$y = ax^2 + bx + c$$

$$= x^2 - 2x - 3$$

c is y-intercept

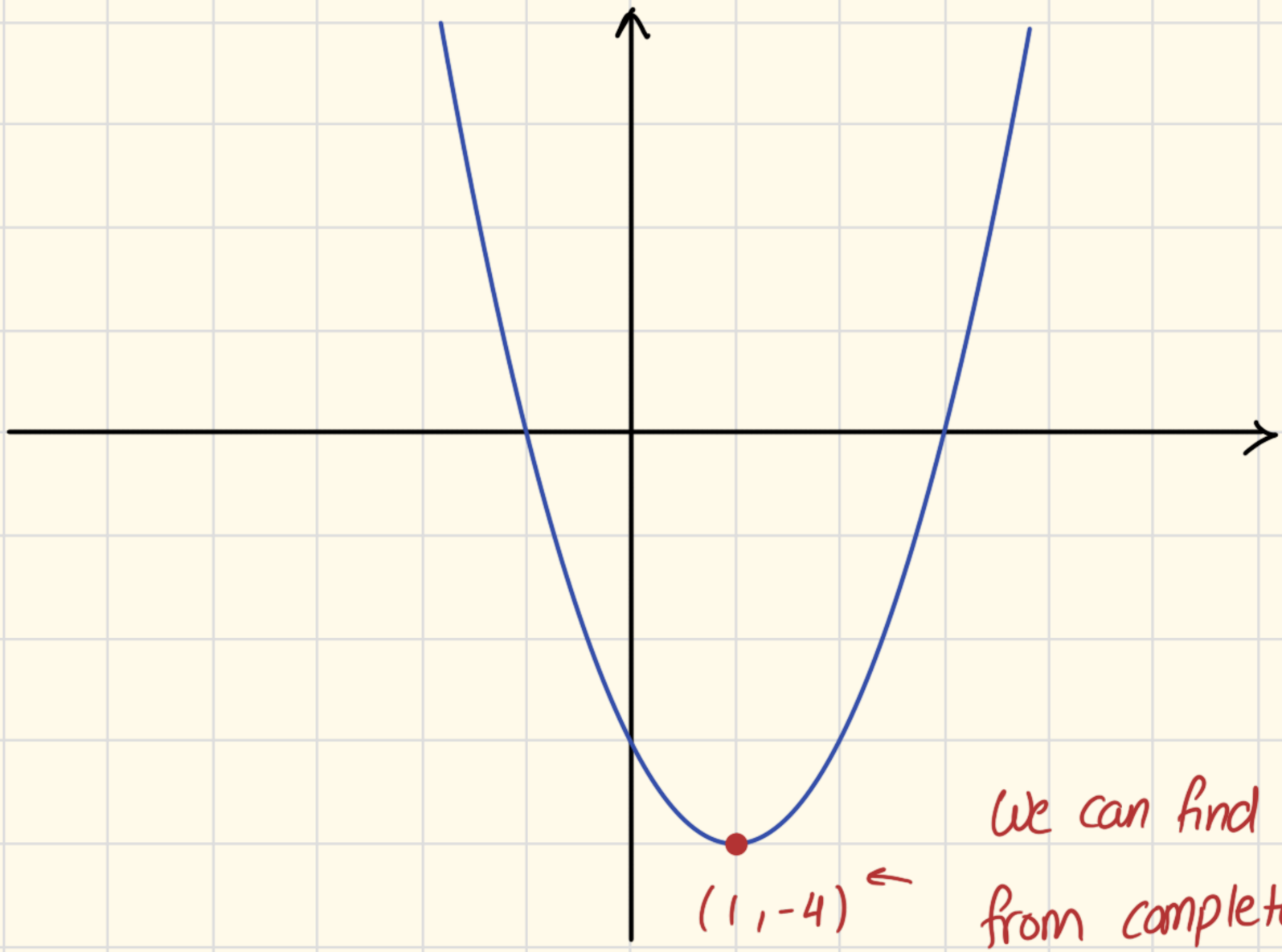


QUADRATIC GRAPHS

$$y = (x-h)^2 + K$$

$$= (x-1)^2 - 4$$

(h, K) are coordinates of min OR max



(1, -4) ←

We can find min from complete square form